Hexagon Number Pyramids

Year 4 · Supplementary Worksheet — extra practice & worked breakdowns

Worked Examples

Example 1 — Building a 3-hexagon pyramid: A 3-hexagon base row has 5, 8, 3. What is the top number?

Answer: Row 2: 5+8=13, 8+3=11. Top row: 13+11=24.

Example 2 — Building a 4-hexagon pyramid: A 4-hexagon base row has 2, 7, 4, 6. What is the top number?

Answer: Row 2: 2+7=9, 7+4=11, 4+6=10. Row 3: 9+11=20, 11+10=21. Top: 20+21=41.

Example 3 — Finding a missing base number: A 3-hexagon base row has 4, 10 and an unknown third number. The top hexagon is 36. Find the missing base number.

Answer: Let the missing number be x. Row 2: 4+10=14, 10+x. Top = 14+(10+x) = 24+x = 36, so x = 12.

Example 4 — Working with a 5-hexagon pyramid: A 5-hexagon base row has 2, 3, 4, 5, 6. What is the top number?

Answer: Row 2: 5,7,9,11. Row 3: 12,16,20. Row 4: 28,36. Top: 64.

Example 5 — Doubling the base: A 3-hexagon base row has 3, 4, 5, giving a top number of 16. If every base number is doubled (6, 8, 10), what is the new top number?

Answer: Doubling every base number doubles every sum built from them. Check by rebuilding: row 2 becomes 6+8=14 and 8+10=18; top row is 14+18=32, which is exactly 16 × 2. New top number: 32.

Foundation

(15 questions)

1. A 3-hexagon base row has the numbers 2, 15, 5. Each hexagon above is the sum of the two hexagons below it. What is the top number?
2. A 3-hexagon base row has the numbers 13, 14, 5. Each hexagon above is the sum of the two hexagons below it. What is the top number?
3. A 3-hexagon base row has the numbers 8, 7, 2. Each hexagon above is the sum of the two hexagons below it. What is the top number?

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4. A 3-hexagon base row has the numbers 13, 4, 14. Each hexagon above is the sum of the two hexagons below it. What is the top number?
5. A 3-hexagon base row has the numbers 15, 2, 4. Each hexagon above is the sum of the two hexagons below it. What is the top number?
6. A 3-hexagon base row has the numbers 3, 4, 3. Each hexagon above is the sum of the two hexagons below it. What is the top number?
7. A 3-hexagon base row has the numbers 3, 12, 14. Each hexagon above is the sum of the two hexagons below it. What is the top number?
8. A 3-hexagon base row has the numbers 7, 7, 12. Each hexagon above is the sum of the two hexagons below it. What is the top number?
9. A 3-hexagon base row has the numbers 11, 6, 8. Each hexagon above is the sum of the two hexagons below it. What is the top number?
10. A 3-hexagon base row has the numbers 7, 11, 4. Each hexagon above is the sum of the two hexagons below it. What is the top number?
11. A 3-hexagon base row has the numbers 12, 6, 8. Each hexagon above is the sum of the two hexagons below it. What is the top number?
12. A 3-hexagon base row has the numbers 2, 9, 9. Each hexagon above is the sum of the two hexagons below it. What is the top number?
13. A 3-hexagon base row has the numbers 10, 5, 6. Each hexagon above is the sum of the two hexagons below it. What is the top number?
14. A 3-hexagon base row has the numbers 4, 15, 11. Each hexagon above is the sum of the two hexagons below it. What is the top number?
15. A 3-hexagon base row has the numbers 2, 3, 2. Each hexagon above is the sum of the two hexagons below it. What is the top number?

Standard

(20 questions)

1. A 4-hexagon base row has 6, 8, 11, 20 (each hexagon above = sum of the two below). What is the top number?
2. A 4-hexagon base row has 16, 13, 5, 14 (each hexagon above = sum of the two below). What is the top number?
3. A 4-hexagon base row has 20, 9, 6, 4 (each hexagon above = sum of the two below). What is the top number?
4. A 4-hexagon base row is 17, ?, 18, 7. The second row up reads 37, 38, 25 and the top hexagon is 138 (addition rule). What is the missing base number?
5. A 4-hexagon base row has 10, 7, 8, 19 (each hexagon above = sum of the two below). What is the top number?
6. A 4-hexagon base row has 8, 4, 20, 15 (each hexagon above = sum of the two below). What is the top number?

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7. A 4-hexagon base row has 4, 7, 17, 8 (each hexagon above = sum of the two below). What is the top number?
8. A 4-hexagon base row is 9, ?, 5, 4. The second row up reads 26, 22, 9 and the top hexagon is 79 (addition rule). What is the missing base number?
9. A 4-hexagon base row has 7, 5, 3, 7 (each hexagon above = sum of the two below). What is the top number?
10. A 4-hexagon base row has 10, 6, 5, 18 (each hexagon above = sum of the two below). What is the top number?
11. A 4-hexagon base row has 20, 16, 20, 18 (each hexagon above = sum of the two below). What is the top number?
12. A 4-hexagon base row is 20, ?, 16, 12. The second row up reads 29, 25, 28 and the top hexagon is 107 (addition rule). What is the missing base number?
13. A 4-hexagon base row has 3, 3, 18, 19 (each hexagon above = sum of the two below). What is the top number?
14. A 4-hexagon base row has 19, 3, 3, 11 (each hexagon above = sum of the two below). What is the top number?
15. A 4-hexagon base row has 3, 9, 18, 17 (each hexagon above = sum of the two below). What is the top number?
16. A 4-hexagon base row is 6, ?, 9, 10. The second row up reads 11, 14, 19 and the top hexagon is 58 (addition rule). What is the missing base number?
17. A 4-hexagon base row has 5, 12, 6, 8 (each hexagon above = sum of the two below). What is the top number?
18. A 4-hexagon base row has 5, 3, 16, 9 (each hexagon above = sum of the two below). What is the top number?
19. A 4-hexagon base row has 8, 15, 13, 20 (each hexagon above = sum of the two below). What is the top number?
20. A 4-hexagon base row is 8, ?, 5, 8. The second row up reads 18, 15, 13 and the top hexagon is 61 (addition rule). What is the missing base number?

Hard

(5 questions)

1. A 5-hexagon base row has 10, 11, 11, 22, 6. Each hexagon above is the sum of the two below. What is the single top number?
2. A 3-hexagon base row has 11, 6 and an unknown third number. The top hexagon (sum of all three combined through the addition rule) is 26. Find the missing base number.
3. Build the full pyramid from base row 10, 10, 18 (addition rule). What is the sum of every number in the whole pyramid (all rows added together)?

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4. A 4-hexagon base row has 5, 9, 2, 3. If every base number were doubled, what would the new top number be?
5. A 2-hexagon base row has 23 and an unknown number. The hexagon above them (their sum) is 97. What is the missing number?

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